MILITARY SPECIFICATION

RADIO SET AN/GRC-103(V)1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the following units of a radio relay equipment in the frequency range 220 to 404.5 MHz and designated Radio Set AN/GRC-103(V)1.

Receiver, Radio R-1329( )(P)/GRC-103(V)

Amplifier-Converter AM-4316( )/GRC-103(V)

Transmitter, Radio T-983( )(P)/GRC-103(V)

Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V)

Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)

Antenna AS-1852( )/GRC-103(V)

Mast AB-952( )/GRC-103(V)

Cable Assembly, RF CG-3444( )/U (1 ft 6 in)

Cable Assembly, RF CG-3444( )/U (3 ft 6 in)

Cable Assembly, RF CG-3443( )/U (80 ft)

Adaptor, Connector UG-1375( )/U

Cable Assembly, Special Purpose, Electrical, Branched CX-10763( )/GRC-103(V) (3 ft 6 in)

The radio set uses the following special accessories for extension of capabilities.

Extension Kit, Mast MK-1009( )/GRC-103(V)

Converter, Frequency, Electronic CV-2500( )/GRC

Cable Assembly, Special Purpose, Electrical CX-10879( )/GRC-103(V)

Dummy Load, Electrical DA-437( )/GRC-103(V)

Cable Assembly, Special Purpose, Electrical CX-10762( )/U (5 ft)

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shock tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirement for (Navy)
MIL-P-11268	Parts, Materials and Processes used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-R-45292	Radio Set AN/GRC-103(V) (Units of), Packaging and Packing of

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment: Development and Production
MIL-STD-810	Environmental Test Methods

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DRAWINGS

USAECOM

SC-A-46439	List of Accessories for Package Tester
SC-DL-424153	Adapter, Connector UG-1375( )/U
DL-SM-B-696006	Receiver, Radio R-1329( )(P)/GRC-103(V)
SM-A-696070	Test Procedure, Mast AB-952( )/GRC-103(V)
DL-SM-B-696151	Mast AB-952( )/GRC-103(V)
DL-SM-B-696171	Transmitter, Radio T-983( )(P)/GRC-103(V)
DL-SM-B-696175	Recv Trans, Order Wire, RT-773( )/GRC-103(V)
DL-SM-B-696203	Cable Assy, R.F. CG-3443( )/U
DL-SM-B-696207	Cable Assy, R.F. CG-3444( )/U
DL-SM-B-696210	Cable Assy, S.P. Elec., CX-10763( )/GRC-103(V)
SM-A-696443	Test Specification - Amplifier-Frequency Multiplier
SM-A-698002	Test Specification - Intermediate Frequency Amplifier
SM-A-698031	Test Specification - Electrical Frequency Limiter-Discriminator
SM-A-698035	Test Specification - Video Amplifier
SM-A-698241	Test Specification - Pulse Form Restorer
SM-A-698317	Test Specification - Panel Monitor
SM-A-698363	Test Specification - Power Supply
SM-A-698372	Test Specification - Power Supply
SM-A-698610	Test Specification - Amplifier-Frequency Multiplier
SM-A-698628	Test Specification - Electrical Frequency Synthesizer
SM-A-698684	Test Specification - Amplifier Monitor
SM-A-698685	Test Specification - Distribution Box
SM-A-698695	Test Specification - Monitor Panel
SM-A-698759	Test Specification - Distribution Panel
SM-A-698776	Test Specification - Alarm Control
SM-A-698789	Test Specification - Distribution Box
SM-A-698790	Test Specification - Monitor Panel
SM-A-698805	Test Specification - Telephone Signal Converter
SM-A-698806	Test Specification - Amplifier Assembly
SM-A-698807	Test Specification - RT-773( )/GRC-103(V)
SM-A-698826	Test Specification - Radio Receiver R-1329( )(P)/ GRC-103(V)
SM-A-698831	Test Specification - Transmitter T-983( )(P)/ GRC-103(V)
DL-SM-B-763660	Amplifier, Converter AM-4316( )/GRC-103(V)
DL-SM-B-763661	Amplifier, Frequency Multiplier AM-4320( )/ GRC-103(V)

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DRAWINGS - (Contd)

DL-SM-B-763662	Antenna AS-1852( )/GRC-103(V)
SM-A-763665	Antenna AS-1852( )/GRC-103(V)
SM-A-763688	Test Specification - Control-Monitor, Signal Level
SM-A-763695	Test Specification - Converter, Frequency, Electronic
SM-A-763697	Test Specification - Amplifier, Radio Frequency
SM-A-763698	Test Specification - Tuner, Radio Frequency
SM-A-763699	Test Specification - Control-Indicator
SM-A-763700	Test Specification - Amplifier-Converter AM-4316( )/GRC-103(V)
SM-A-763730	Test Specification - Electronic Switch
SM-A-763731	Test Specification - Frequency Multiplier
SM-A-763733	Test Specification - Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V)
SM-A-763738	Test Specification - AN/GRC-103, Receiver Section
SM-A-763739	Test Specification - AN/GRC-103, Transmitter Section
SM-A-763740	Test Specification - AN/GRC-103, Terminal Section
SM-A-764122	Test Specification - Amplifier, Radio Frequency
SM-A-764157	Test Specification - Duplexer

3. REQUIREMENTS

3.1 First Article - When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction - The equipment shall be constructed in accordance with the requirements of this specification and the drawing and data list as indicated, including the requirements for parts, materials and processes thereon.

DL-SM-B-696006	Receiver, Radio R-1329( )(P)/GRC-103(V)
DL-SM-B-763660	Amplifier-Converter AM-4316( )/GRC-103(V)
DL-SM-B-696171	Transmitter, Radio T-983( )(P)/GRC-103(V)
DL-SM-B-763661	Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V)
DL-SM-B-696175	Recv Trans, Order Wire RT-773( )/GRC-103(V)
DL-SM-B-763662	Antenna AS-1852( )/GRC-103(V)
DL-SM-B-696151	Mast AB-952( )/GRC-103(V)
DL-SM-B-696207	Cable Assy, R.F. CG-3444( )/U
DL-SM-B-696203	Cable Assy, R.F. CG-3443( )/U
DL-SM-B-696210	Cable Assy, S.P. Elec, CX-10763( )/GRC-103(V)
SC-DL-424153	Adapter, Connector UG-1375( )/U

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3.3 Parts, materials and processes; general - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.4 Electrical characteristics - Radio Set AN/GRC-103(V)1 provides for multichannel, line-of-sight, two-way communication in the frequency range of 220-404.5 MHz (Channels 40-409). This version is intended primarily for use as a radio link in a communication network, which may include carrier telephone and teletypewriter sets. The radio set is used in pairs; one at each end of the line-of-sight path. The radio set may be used with 6/12 or 24 channel pulse code modulation (PCM) equipment or with 4 channel frequency division modulation (FDM) when associated with appropriate interface equipment. The system shall provide channel assignments on each half and full megacycle points (1/2 MHz spacing).

3.4.1 Transmitter section -

3.4.1.1 Output power - The normal transmitter RF output power, without the RF Duplexer, shall be not less than 30 watts at any frequency in the range of 220 to 404.5 MHz.

3.4.1.2 Low power alarm - A visual and audible alarm is provided in the transmitter. The visual alarm shall indicate any change in RF output power that is greater than 8 dB ± 2 dB below the nominal power output rating of the transmitter. The switch associated with the audible alarm is included such that: In one switch position, the alarm shall sound when the visual alarms are lighted as in the case of a failure; in the second switch position, the alarm sounds when the light is not lighted. The audible alarm thus indicates a change in state of the visual signal.

3.4.1.3 Output tube dissipation - The transmitter output power shall not decrease below specification limits after:

a. Running the transmitter for 15 minutes with its output terminals short-circuited.

b. Running the transmitter for 15 minutes with its output terminals open-circuited.

c. Running the transmitter for 30 minutes without drive.

3.4.1.4 Frequency setability - It shall be possible to set the transmitter to any desired frequency at 0.5 MHz intervals in the frequency range of 220 to 404.5 MHz, with an accuracy of better than 0.002% corresponding to 20 Hz per MHz.

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3.4.1.5 Transmitter frequency response - The overall frequency response of the transmitter when a modulation signal to produce 100 kHz peak deviation is applied to the video input shall conform to Table I.

TABLE I

RELATIVE ATTENUATION (db) REF 10 kHz			
FREQUENCY kHz	OPTIMUM	MINIMUM	MAXIMUM
.125	1.0	0.5	1.5
.200	.5	0	1.0
.250	.2	0	.4
1.0	0	-0.2	0.2
10	0(REF)	--	--
30	0	-0.2	.2
60	.1	0	.3
120	.15	0	.5
240	.7	.5	1.3
360	1.5	0.6	2.3
480	2.7	1.9	4.2
600	4.2	3.4	6.2
720	6.0	5.3	8.8
840	8.2	7.5	11.5
960	10.7	8.5	15.0

3.4.1.6 Spurious outputs - When measured with duplexer, the power output of any harmonic shall be at least 80 dB below that of the fundamental and the power output of any non-harmonic extraneous emission of a band outside plus and minus 15 MHz of the carrier shall be at least 80 dB below that of the fundamental. All spurious outputs over the frequency range $f_o \pm 15$ MHz shall be suppressed at least 60 dB, except for the range occupied by the modulation, where the spurious outputs shall not exceed the level of the modulation envelope.

3.4.1.7 FDM modulation requirements - With a 1 KHz sinusoidal signal of 191 MV applied to the transmitter video input, and the input control adjusted to give a reading of 50% on the panel meter, the transmitter peak deviation shall be between 65 kHz and 97 kHz. This requirement shall be met with the meter switch in the FDM position.

3.4.2 Receiver section - The following requirements shall be met, when tested with the duplexer.

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3.4.2.1 Frequency response shaping - The overall frequency response of the receiver when a frequency modulated signal of constant deviation (100 kHz peak) is applied to the receiver antenna terminal and measured at the video output terminal shall conform to Table II.

TABLE II

RELATIVE ATTENUATION (db) REF 10 kHz			
FREQUENCY kHz	OPTIMUM	MINIMUM	MAXIMUM
0.125	1.0	-0.2	2.5
0.200	0.5	-0.2	1.0
0.250	0.2	-0.2	0.4
1	0	-0.1	0.1
10	0(REF)	--	--
30	0	-0.1	0.1
60	0.15	0.05	0.25
120	0.6	0.2	0.9
180	1.3	0.8	1.8
240	2.3	1.8	2.8
360	5.1	4.4	5.8
480	9.0	8.2	10.0
600	14.0	12.5	15.5
720	20.2	18.4	22.6
840	27.5	25.3	31.0
960	36.0	33.0	40.0

3.4.2.2 Noise figure - The noise figure of the receiver, including the losses due to the receiver protection circuit and the duplexer, shall not exceed 8 dB between 220 and 399.5 MHz and 9 dB between 400 MHz and 404.5 MHz.

3.4.2.3 Demodulation - For received signals deviated ± 300 kHz, ± 75 kHz, and ± 45 kHz, the video outputs shall be 0.265V ± 0.03 V RMS at the VID Connector, 0.189 V ± 0.02 V RMS at the FDM Connector and -10 dBm ± 2 dB (600 OHMS) at the O W Connector, respectively.

3.4.2.4 Received signal monitoring - A panel meter deflection of 60% or more shall be obtained at a received signal level of -50 dBm, and 10% or more at a level of -95 dBm.

3.4.2.5 IF bandwidth - The IF selectivity shall have a nominal 3 dB bandwidth of 750 kHz with the response down at least 60 dB at ± 2.1 MHz or more from resonance.

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3.4.2.6 Receiver protection - When the transmitter has been tuned for optimum output, it shall be possible to tune the receiver duplexer section through the transmitter frequency without damage to the receiver. The circuitry shall provide protection against external signals at any frequency with power level up to +47 dBm under all conditions of receiver tuning. Protection shall be provided against pulse modulated signals of any frequency of 1 Kw peak power, duration 1 us, duty cycle 0.001 and a rise time of 10 ns from 10% to 90% amplitude.

3.4.2.7 Spurious response - The ratio of desired signal response to any spurious response shall be as follows:

a. For a maximum of 6 discrete responses, the ratio shall be at least 80 dB.

b. For all other responses, the ratio shall be at least 100 dB.

3.4.3 Terminal section (system requirements) -

3.4.3.1 Input Impedance - The impedance at the video input to the transmitter shall be 91 ohms ± 10% resistive to at least 480 kHz and 91 ohms ± 15% to at least 800 kHz.

3.4.3.2 Output Impedance - The output impedance of the receiver shall be 51 ohms ± 10% resistive for the 24 channel PCM output to at least 800 kHz and, for the regenerated 6/12 channel PCM output such that when terminated in a resistive load of 91 ohms ± 5%, the waveform is as specified in 3.4.3.7.1.

NOTE: All received RF signal levels are measured at the antenna terminals of the duplexer.

3.4.3.3 Frequency response (PCM) - The overall frequency response between transmitter video input and receiver video output relative to the response at 10 kHz shall conform to Table III.

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TABLE III

RELATIVE ATTENUATION (dB) REF 10 kHz			
FREQUENCY kHz	NOMINAL	MINIMUM	MAXIMUM
.125	2.0	1.0	3.0
.200	1.0	0	2.0
.250	.4	0	.8
.500	0	-.1	+.1
1.0	0	-.1	+.1
10	0(REF)	--	--
30	0	-.1	.2
60	.2	.1	.4
120	.7	.5	1.0
180	1.6	1.3	2.2
240	2.9	2.5	4
360	6.5	5.7	8
480	11.7	10.2	13.5
600	18.2	16.0	21.0
720	26.2	23.1	30.5
840	35.7	31.7	41.0
960	46.7	41.2	52.5

3.4.3.4 PCM modulation requirements - When the modulation sensitivity is adjusted to maximum, an input voltage of 1 volts peak-to-peak shall deviate the transmitter plus and minus 300 kHz peak. Also the metering circuitry shall enable the operator to adjust the peak deviation of the transmitter to plus and minus 180 kHz and plus and minus 300 kHz with an accuracy of plus 2.2 and minus 1.25 dB with inputs between 1 volt to 4 volts peak-to-peak. The transmitter shall be considered to have met this requirement if when tested in accordance with the 12 and 24 ch PCM metering sections of the test specified in 4.19.3 it meets the specifications thereof.

3.4.3.5 Signal to noise (PCM) - The transmitter shall be tuned to any of its operating frequencies and shall be operated at rated RF power output. A 10 kHz sinusoidal signal shall be applied at the video input terminals of the transmitter and the transmitter shall be adjusted for a peak deviation of plus and minus 180 kHz. The receiver output, when the receiver input is coupled to the transmitter through sufficient attenuation, shall be 0.45V $\pm$ 0.15V peak to peak at the unregenerated output terminals.

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Under these conditions, the ratio of average detected signal power to the average noise power at the unregenerated video output terminals of the receiver as measured with a flat weighted meter shall be at least 12 dB when the receiver RF signal is -94 dBm and at least 31 dB when the received RF signal is -74 dBm. The transmitter shall be adjusted for rated peak deviation (plus and minus 300 kHz) and the receiver unregenerated video output shall be 3/4 ± 1/4 volt peak-to-peak, the ratio shall be at least 16 dB when the received RF signal is -94 dBm and at least 35 dB when the received signal is -74 dBm.

3.4.3.6 Distortion (PCM)- The equipment shall be arranged as in 3.4.3.5 with the receiver coupled to the transmitter through sufficient attenuation so that the RF input at the receiver input terminals (including duplexer) is minus 54 dBm. A sinusoidal signal at any frequency within the range of 250 Hz to 100 kHz shall be applied at the video input terminals of the transmitter at a level of 2 volts peak-to-peak. The transmitter shall be adjusted for rated deviation and the receiver output shall be 3/4 plus or minus 1/4 volt peak-to-peak at the unregenerated video output terminals. Under these conditions, the total harmonic distortion measured at the unregenerated video output terminals shall not exceed a level 26 dB below the fundamental.

3.4.3.7 Pulse regeneration (PCM) - The receiver output circuits shall be capable of generating new pulses whose amplitude, timing and pulse shape are derived from the 6 channel or 12 channel signals briefly described below.

PCM CHARACTERISTICS

Number of channels	6	12
Signal Form	(Full-band, Unipolar Binary)	
Bit Rate	288	576
Pulse Width at Half Amplitude	3.472 µsec	1.736 µsec

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3.4.3.7.1 Regenerated PCM - The regeneration shall eliminate all noise peaks less than 40% of the pulse amplitude for the range of unregenerated video output levels corresponding to input deviations of ± 145 peak minimum to ± 230 kHz peak maximum. The regenerated signal amplitude shall be -0.1 volt ± 0.1 volt for the '1' condition and -2.0 volts ± 0.20 volts for the '0' condition, delivered into a load of 91 ohms ± 5%, with no reversal of phase from the transmitter input or unregenerated video output. Pulse width shall be full interval with 10% to 90% rise time 10 nsec minimum, 50 nsec maximum. The 50% point shall be 0 nsec minimum, 80 nsec maximum after the 50% point of the rising edge of the timing signal specified in 3.4.3.7.2.

3.4.3.7.2 Timing Signal - Timing signals shall be extracted from the incoming pulse train and shall accompany the regenerated PCM to the next unit. The timing signal amplitude shall be 2 volts ± 0.2V DC isolated, delivered into a load of 91 ohms ± 5%. Pulse width at the 50% amplitude point shall be 80 nsec minimum, 150 nsec maximum. Rise time of the leading edge from the 10% point to the 90% point shall be 20 nsec minimum, 50 nsec maximum. Pulse rate shall be 576 kHz for both the 6 channel and 12 channel signals.

3.4.3.7.3 Squelch and Alarm - The squelch shall disable the regenerated PCM signal when the signal-to-noise ratio at the unregenerated video output is less than 7 dB ± 2 dB when measured as in 3.4.3.5, with a reference deviation of ± 180 kHz peak. The audible and visible alarm shall operate in conjunction with the squelch to indicate locally, failure of the received signal from the remote station. A reversing switch shall be provided on the front panel such that regardless of the level of signal input to the receiver, the audible alarm can be disabled. The sensitivity of the squelch circuit shall be such that the point of operation which indicates a received signal (lamp extinguished) shall be -98 dBm ± 3 dB.

3.4.3.8 FDM Demodulation Requirements - With a signal deviated 75 kHz applied to the receiver antenna terminal, the level measured across the video output connector terminated in 91 ohms shall be .189 V ± .02 V RMS. The meter of the receiver shall be in the green area with the switch in the FDM position.

3.4.3.9 Duplexers - Insertion loss at the center frequency of the pass-band of the transmitter section shall not exceed 2 dB over 90% of the operating band and shall not exceed 2.5 dB over the remainder of the operating band. Insertion loss at the center frequency of the band-pass of the receiver section shall not exceed 3 dB.

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3.4.3.10 Duplexer discrimination - The discrimination of the filter for the transmitter section shall be a minimum of 25 dB at ± 16 MHz and for the receiver section a minimum of 60 dB at ±16 MHz and 100 dB at ±50 MHz.

3.4.3.11 Voltage standing wave ratio - The voltage standing wave ratio measured at the antenna terminal of the duplexer, with the transmitter terminal terminated in a 50 ohm load and the receiver connected to the receiver connection, shall not exceed 2.2:1 at any transmitted frequency within the operating frequency range as specified in 3.4.

3.4.3.12 Systems level stability - The transmitter and receiver shall be connected through an attenuator with sufficient attenuation such that the power input to the receiver input terminals is minus 60 dBm. With the transmitter modulated as described in 3.4.3.4 (PDM modulation requirements) and the overall system adjusted for nominal gain, the following conditions shall be met: The system gain shall not vary more than ± 1 dB in any eight hour period between the 91 ohm video input and 51 ohm video output terminals of the receiver when the received signal is varied between -95 dBm and -45 dBm under any combination of temperature, humidity or line voltage variation as specified herein.

3.4.4 Order wire -

3.4.4.1 Order wire operation - The order wire unit shall meet the following requirements to be compatible with Multiplexers TD-202( )/G, TD-204( )/G, and TD-754( )/G.

3.4.4.1.1 Tone generator - When the tome generator is activated, the level measured at the TRANS "A" and TRANS "B" terminals of the order units shall be -10 dBm ± 1 dBm.

3.4.4.1.2 Signal receiver detector sensitivity - When a -10 dBm ± 6 dB, 1600 Hz ± 4 Hz signal is applied to either the REC A or REC B terminals of the order wire unit, the signal lamp shall light and the buzzer sound.

3.4.4.2 Receiver demodulation order wire - With an RF signal deviated ± 45 kHz peak applied to the receiver antenna terminal, the level measured at the receiver order wire shall be -10 dBm ± 2 dB.

3.4.4.3 Transmitter deviation order wire -A 1000 Hz sinusoidal signal at a level of zero dBm applied to the microphone input terminals of the order wire unit, shall produce a peak deviation of the transmitter between 38 kHz and 59 kHz. A 1000 Hz signal of -10 dBm applied to the transmitter order wire connector shall produce a peak deviation of between 39 kHz and 58 kHz.

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3.4.4.4 Signal-to-noise (order wire) - The transmitter and receiver shall be connected together in a systems arrangement with sufficient attenuation between them such that the power measured at the input terminals of the receiver is -94 dBm. With the transmitter modulated as in 3.4.4.3 the signal-to-noise ratio measured with F1A weighting at the local order wire terminals (receive phone terminals) shall be at least 29 dB at any operating frequency in the presence of a normal 6/12 channel PCM signal and at least 40 dB in the FDM mode of operation. When measured with C-message weighting the signal-to-noise ratio shall be at least 27.5 dB and 38.5 dB, respectively.

3.4.4.5 Peak limiting (order wire) - The limiting action to be provided in the transmitter for the order wire shall be such as to compress test tone power inserted at the microphone terminals in accordance with Table IV.

TABLE IV

Limiting Characteristic, Insert Path

Measured between MIKE terminal and TRAN 'A' or TRAN 'B' terminals

INPUT POWER	COMPRESSION
-10 dBm	None
-5 dBm	None
0 dBm	Less than 0.5 dB
+5 dBm	3.5 dB ± 1 dB
+10 dBm	7.5 dB ± 2 dB
+15 dBm	10.0 dB ± 3 dB

3.4.4.6 Frequency response (order wire system) - The overall frequency response characteristic between the local microphone input and the phone output terminals of the order wire unit at a distant station shall be plus 1 to minus 3 dB from 300 to 1800 Hz, referred to the 1 kHz response at an RMS level of zero dBm, when the equipment is employed in a full duplex PCM system.

3.4.4.7 Frequency response (order wire unit) - The order wire unit RT-773 ( )/GRC-103 shall meet the following frequency response characteristics (Table V) when measured with input levels specified in 3.4.4.11.

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TABLE V

OW FREQUENCY RESPONSE

PATH	FROM	TO	FREQUENCY RANGE	RESPONSE REF: 1 kHz
'THRU'	REC 'A'	TRAN 'B'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
'THRU'	REC 'B'	TRAN 'A'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
DROP	REC 'A'	PHONE terminal	200 Hz -2.0 kHz	+0.5 dB -1.5 dB
DROP	REC 'B'	PHONE terminal	200 Hz -2.0 kHz	+0.5 dB -1.5 dB
INSERT	MIKE terminal	TRAN 'A'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
INSERT	MIKE terminal	TRAN 'B'	200 Hz -2.0 kHz	+0.5 dB -1.0 dB
SIDETONE	MIKE terminal	PHONE terminal	200 Hz -2.0 kHz	± 1.0 dB

3.4.4.8 Distortion (order wire system) - With the equipment arranged as in 3.4.4.4, the total harmonic distortion measured at the order wire output terminals shall not exceed 5%.

3.4.4.9 Distortion (order wire units) - The distortion of the order wire unit when measured with a 1000 Hz signal as specified in 3.4.4.7 shall be as follows:

'THRU' PATHS	REC 'A' to TRAN 'B'	Less than 1%
	REC 'B' to TRAN 'A'	Less than 1%
INSERT PATHS	MIKE terminal to TRAN 'A'	Less than 2%
	MIKE terminal to TRAN 'B'	Less than 2%
DROP PATHS	REC 'A' to PHONE terminal	Less than 1%
	REC 'B' to PHONE terminal	Less than 1%

3.4.4.10 Net loss (order wire system) - The net loss between the local microphone input and the phone output at a distant station shall be 18 dB ± 3 dB.

3.4.4.11 Order wire unit levels - The order wire unit levels when measured with a 1000 Hz signal shall conform to Table VI:

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TABLE VI

FROM	TO	OUTPUT LEVEL	INPUT LEVELS
REC 'A'	TRAN''B'	-10 dBm ± 0.1 dB (adjustable)	-10 dBm
REC 'B'	TRAN 'A'	-10 dBm ± 0.1 dB (adjustable)	-10 dBm
REC 'A'	PHONE terminal	-18 dBm ± 1.0 dB	-10 dBm
REC 'B'	PHONE terminal	-18 dBm ± 1.0 dB	-10 dBm
MIKE terminal	TRAN 'A'	-10 dBm ± 1.0 dB	0 dBm
MIKE terminal	TRAN 'B'	-10 dBm ± 1.0 dB	0 dBm
MIKE terminal	PHONE terminal	-24 dBm ± 2.0 dB (sidetone)	0 dBm
REC 'A'	TRAN 'A'	-55 dBm ± 10.0 dB (crosstalk)	-10 dBm
REC 'B'	TRAN 'B'	-55 dBm ± 10.0 dB (crosstalk)	-10 dBm

3.4.5 Antenna -

3.4.5.1 Symmetry - The ratio of the half-power (-3 dB) E- and H- plane beamwidth shall not exceed 2.0:1 over the frequency range.

3.4.5.2 Squint - The gain of the antenna measured in the direction of the normal from the antenna shall not differ from the maximum gain measured at any frequency in the operating range by more than 0.5 dB.

3.4.5.3 Minor-lobes - The ratio of the major lobe of the antenna to any minor lobe shall be at least 12 dB at any frequency in the operating range.

3.4.5.4 Gain - The gain of the antenna relative to an isotropic radiator shall be at least 9.2 dB at 220 MHz, 9.3 dB at 250 MHz, 9.7 dB at 300 MHz, 10.6 dB at 350 MHz, 12.2 dB at 400 MHz, and 12.3 dB at 405 MHz. At intermediate frequencies, the gain shall exceed the values linearly interpolated between the frequencies specified.

3.4.5.5 Voltage standing wave ratio - The VSWR measured at the antenna terminals shall not exceed 1.75:1 over the operating frequency range.

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3.4.5.6 Impedance - The nominal input impedance of each antenna shall be 50.0 ohms.

3.5 Mast -

3.5.1 Erection - The mast shall be capable of erection with the antenna and transmission line installed, in winds of twenty-five miles per hour with the launcher guyed and all other guy wires slack.

3.5.2 Operation - The mast assembly erected with guy wires tensioned and antenna and transmission line installed, shall withstand the following conditions without permanent deformation, and maintain the antenna azimuth within $\pm 5^{o}$ of any initial setting.

a. Wind velocities up to 75 mph or;

b. Wind velocities up to 50 mph with a one-half inch radial ice load on the entire mast-antenna installation.

3.6 Finish, protective - Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware type items, such as handles, hinges, screws, etc., and necessary touch-up after mounting. Unless otherwise specified, the final paint film on type I surface (see MIL-F-14072), exposed to view during operation, shall be a smooth, semi-gloss (olive drab) green, in color conforming to No. X24087 of FED-STD-595. When some other color or gloss is referred to in the equipment specification, but the details are not specified, the contractor shall apply to the Contracting Officer for information regarding the required finish. For interference reduction purposes, interface surfaces of structures and bonding connections between cases and mounting bases shall have a finish for electrical contact. The top coat of paint on the front panel of the equipment shall be semi-gloss gray enamel in accordance with MIL-E-15090.

3.7 Marking -

3.7.1 General - Marking shall conform to MIL-M-13231. Front panel marking shall be group I as described in that specification.

3.7.2 Visibility - Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the equipment.

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3.7.3 Chassis marking - All chassis identification, parts identification, warning notes, etc., shall be applied by the silk screen process or processes giving the same degree of legibility, placement accuracy and permanence. The marking shall be in accordance with the drawings.

3.7.4 Serial numbers - The following items shall have serial numbers and name plates.

Transmitter, Radio T-983( )P/GRC-103(V)

Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V)

Receiver, Radio R-1329( )P/GRC-103(V)

Amplifier-Converter AM-4316( )/GRC-103(V)

Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)

Antenna AS-1852( )/GRC-103(V)

Mast AB-952( )/GRC-103(V)

3.7.4.1 Composition of serial numbers - The serial numbers used for this equipment shall be ten digits. The first five digits shall be the Federal Identification number of the prime manufacturer. The sixth digit shall be a code letter to indicate the production contract with said manufacturer. A for first production, B for second production, etc. The final four digits shall be progressively from 0000 to 9999 as required. For substantially continuous production on more than one contract, consecutive numbering without change of letter designator may be used upon application to and approval by the Contracting Officer. Records will be maintained and furnished the Government of the first and last serial number of each unit, assembly, subassembly, and module furnished on the contract.

3.8 Power requirements - The equipment shall operate from a 115 Volt rms, 47.5 Hertz to 420 Hertz, single phase source, having a voltage variation of plus or minus 10%. The total power drain for a transmitter and receiver shall not exceed 400 watts at normal operating conditions.

3.9 Electrical Interchangeability - Like units, assemblies, subassemblies, and replaceable parts shall be functionally interchangeable without modification or adjustment except operator adjustments of such items or of the equipment. Units, assemblies, subassemblies and replaceable parts shall meet the electrical requirements of their respective drawings and the tests of 4.19.

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3.10 Mechanical Interchangeability - Like units, assemblies, subassemblies, and replaceable parts shall be physically interchangeable without modification or adjustment except operator adjustments of such items or of the equipment.

3.11 Service conditions -

3.11.1 Temperature -

3.11.1.1 Low - During testing as specified in 4.7.4.1, the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.

3.11.1.2 High - During testing as specified in 4.7.4.2, the equipment shall meet the performance specified in 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1.

3.11.2 Moisture-resistance (humidity) - During testing as specified in 4.7.5, the equipment shall meet the performance requirements of 4.9.1. After testing, the equipment shall meet full specification performance for these measurements as specified in 4.9.1. Upon completion of tests there shall be no evidence of corrosion or degradation of finishes.

3.11.3 Elevation (altitude) - The equipment shall meet full specification performance of 4.9.2 during and after testing as specified in 4.7.6.

3.11.4 Vibration, bounce and shock - The equipment shall be capable of meeting the requirements of Table VII.

TABLE VII

BOUNCE AND SHOCK		
Paragraph heading	Test Para	Performance after test
Bounce	4.7.1	Specified performance. (Notes A & B). No physical damage.
Shock; ballistic	4.7.2	Operable (Note B). Any physical damage shall be minor only.
Shock; bench-handling	4.7.3	Specified performance (Note B). No physical damage.

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NOTE A: With the equipment shock-mounts (if any) blocked, the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz.

NOTE B: The equipment shall meet specified performance for the following:

a. Paragraph 4.2 and 4.5 of SM-A-763739, except test channels shall be 40, 179, 180 and 400 at 115 volts, 60 Hz.

b. Paragraph 4.5 of SM-A-763738, except test channels shall be 43, 124 and 370.

c. Paragraph 4.2.2 of SM-A-763740.

d. Paragraph 4.1 of SM-A-763740, except test channels shall be 40, 179, 180 and 400.

e. Paragraph 4.5.2 of SM-A-763740.

f. Paragraph 4.10 of this specification.

3.12 Electromagnetic interference characteristics - The equipment shall meet the electromagnetic interference requirements of MIL-STD-461, Notice 4, Table A-1 as defined herein. The following Class CE requirements apply to the AN/GRC-103(V)1:

CE06 key-up only	CS01	RE02	RS03.2	CE04B
	CS06	RE02.1		CS02B

3.13 Preconditioning - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the preconditioning of 4.6.

3.14 Workmanship - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

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In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

3.15 Reliability -

3.15.1 Numerical reliability requirement - The equipment shall have a mean-time-between-failure (MTBF) of 1200 hours when operated in accordance with Appendix I to this specification as follows:

a. Temperature: 40° ± 5°C

b. Temperature Cycling: None

c. Vibration: 2.2G ± 10% peak acceleration value at any non-resonant frequency between 20 Hz and 60 Hz, measured at the mounting points of the equipment. The duration of vibration shall be for two minutes of every operating hour during one eight-hour period each week.

d. Equipment On-Off Cycling: Equipment shall be operated continuously for periods of 144 hours and shall be switched off for periods of eight hours.

e. Input Voltage: Nominal Specified Voltage ± 5%.

f. Input Voltage Cycling: None

3.15.2 Reliability program - The contractor shall conduct a reliability program that shall provide for, but shall not be limited to, the following reliability tasks as described in MIL-STD-785:

a. Reliability Management (5.1)

b. Parts Reliability (5.2.3)

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c. Reliability Demonstration (5.3.3)

d. Failure Data (5.4)

3.16 Technical literature, tools, and running spare parts - Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection requirements specified herein are classified as follows:

a. First Article Inspection (see 4.3) - does not include preparation for delivery.

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.5).

4.3 First Article - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First Article Units - The contractor shall furnish seven (7) first article units of the complete Radio Set AN/GRC-103(V)1.

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4.3.2 First Article Inspection - The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B, Group C and Group D (see Table VIII, IX, X and XI, respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units; and designated units for specific subgroups within Group C in accordance with 4.5.3.1, and Group D.

4.3.3 First Article Data - The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspection covered by subsidiary documents - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Parts, Materials, and Processes	3.3
Finish	3.6
Marking	3.7
Workmanship	3.14

4.5 Quality conformance inspection of equipment before preparation for delivery - The contractor shall perform the inspections specified in 4.4, 4.6 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A Inspection - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table VIII. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table VIII.

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TABLE VIII

Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
(a) Visual and Mechanical	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Receiver, Radio (1RE1) R-1329( )(P)/GRC-103(V)				
Synthesizer (1RE1A2)			1%	4%
Panel, Monitor (1RE1A1)			1%	4%
Power Supply (1RE1PS1)			1%	4%
Restorer, Pulse Form (1RE1A3)			1%	4%
Limiter-Discriminator (1RE1A4)			1%	4%
Amplifier-Frequency Multiplier (1RE1A5)			1%	4%
Amplifier, Video (1RE1AR1)			1%	4%
Amplifier, Intermediate Freq(1RE1AR2)			1%	4%
Case, Receiver (1A2) CY-4638/GRC-103(V)			1%	4%
Distribution Box (1A2A1)			1%	4%
Amplifier-Converter(2) AM-4316( )/GRC-103(V)	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Duplexer (2A1A1)			1%	4%
Converter, Frequency, Electronic(2A1A2)			1%	4%
Amplifier, Radio Frequency (2A1AR1)			1%	4%
Control-Indicator (2A2)			1%	4%
Control-Monitor, Signal Level (2A4)			1%	4%
Transmitter, Radio (5TR1) T-983( )(P)/GRC-103(V)	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Synthesizer (5TR1A2)			1%	4%
Panel, Monitor (5TR1A1)			1%	4%
Power Supply (5TR1PS1)			1%	4%
Control, Alarm (5TR1A3)			1%	4%
Amplifier-Frequency Multiplier(5TR1A4)			1%	4%
Amplifier-Monitor (5TR1A5)			1%	4%
Case, Transmitter (5A2)CY-4637( )/GRC-103(V)			1%	4%
Distribution Box (5A2A1)			1%	4%

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Amplifier-Frequency Multiplier (6) AM-4320( )/GRC-103(V)	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Amplifier, Radio Freq (6AR1)			1%	4%
Electronic Switch (6A1)			1%	4%
Frequency Multiplier Assy (6A2)			1%	4%
Control-Indicator (6A3)			1%	4%
Receiver-Transmitter Order Wire (9) RT-773( )/GRC-103(V)	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Panel, Distribution (9A1)			1%	4%
Panel, Monitor (9A2)			1%	4%
Converter, Telephone Signal (9A3)			1%	4%
Amplifier Assembly (9A4)			1%	4%
Case, Recvr-Trans, Order Wire(9MP1)			1%	4%
Antenna (11) AS-1852( )/GRC-103(V)	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Reflector, Antenna (11A1)			1%	4%
Antenna Element (11A2)			1%	4%
Mast (12) (Mast AB-952( )/GRC-103(V))	3.2, 3.3, 3.6, 3.7, 3.14	4.20	1%	4%
Elevator, Antenna			1%	4%
Mast Section			1%	4%
Accessory Kit, Mast			1%	4%
Bag, Mast Accessories			1%	4%
Guy			1%	4%
Ring, Guy Attachment			1%	4%
Spike, Common			1%	4%
Adapter, Antenna			1%	4%
Anchor, Guy			1%	4%
Hammer			1%	4%
Strap Wrench			1%	4%

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Cable Assemblies	3.2, 3.3, 3.6, 3.7, 3.14	4.20		
CG-3444			1%	4%
CG-3443			1%	4%
CX-10763			1%	4%
(b) Electrical (Subassemblies)	3.9	4.19.1		
Receiver, Radio R-1329( )(P)/GRC-103(V)				
Dimmer Switch				
Wideband Noise				
Frequency Deviation				
FDM Output				
OW Output				
PCM Regeneration			1% for the group	
PCM Squelch				
PCM Mute				
Receiver Squelch				
Receiver Signal Monitor				
FDM Return Loss				
Video Return Loss				
Frequency Generation				
Power Output				
Video Frequency Response				
Amplifier-Converter AM-4316/GRC-103(V)	3.9	4.19.2		
High Signal Alarm & Noise Output			1%	
Protection Switch			1%	
Transmitter, Radio T-983( )(P)/GRC-103(V)	3.9	4.19.3		
Voltage Stability				
Video Input and Deviation Metering				
Power Output			1% for the group	
Overheat Alarm				
Low Power				
Output Frequency				
Amplitude-Frequency Response				
Video Input Match				

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Amplifier Frequency Multiplier AM-4320	3.9	4.19.4		
Cathode and Heater Voltage Frequency Selection Power Output and Tracking			1% for the group	
Receiver, Radio (1RE1)				
Synthesizer (1RE1A2)	3.9	4.19.5		
Frequency Check Synch. Alarm Deviation Power Output			1% for the group	
Panel, Monitor (1RE1A1)				
Continuity Check	3.9	4.19.8	1%	
Power Supply (1RE1PS1)	3.9	4.19.9		
Initial Calibration Meter Check Regulation Load & Regulation Unregulated Supply Ripple Measurement (Unreg) Ripple Measurement (Reg) High Freq. Instability			1% for the group	
Restorer, Pulse Form (1RE1A3)	3.9	4.19.6		
Timing Pulse PCM Regenerator OW Noise Test OW Distortion			1% for the group	
Limiter-Discriminator (1RE1A4)	3.9	4.19.7		
Demod. Linearity Deviation Sensitivity Frequency Response			1% for the group	

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Amplifier-Frequency Multiplier (1RE1A5)	3.9	4.19.10		
Harmonic Level			1%	
Amplifier, Video (1RE1AR1)	3.9	4.19.11		
Gain 24CH PCM Adjust Low Signal Alarm Frequency Alarm Distortion Continuity Check			1% for the group	
Amplifier, Intermediate Frequency (1RE1AR2)	3.9	4.19.12		
Gain (without AGC) Frequency Response AGC Limited Output Level			1% for the group	
Case, Receiver (1A2) CY-4638( )/GRC-103(V)			1%	
Distribution Box (1A2A1) Continuity Check	3.9	4.19.13		
Amplifier-Converter (2) AM-4316( )/GRC-103(V) Tuner, Radio Frequency (2A1)	3.9	4.19.14	1%	
Duplexer (2A1A1) Adjustment of Lead Lengths Receiver Channel Calibration Transmitter Channel Calibration	3.9	4.19.15	1% for the group	
VSWR Bandwidth Receiver Path Insertion Loss Receiver Path Ripple Transmitter Path Insertion Loss Transmitter Path Ripple Local Oscillator Signal Tracking			1% for the group	

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Converter, Frequency Electronic (2A1A2)	3.9	4.19.16		
Frequency Tracking VSWR Local Oscillator Calibration Noise Figure			1% for the group	
Amplifier, Radio Frequency (2A1AR1)	3.9	4.19.17		
Gain and Bandwidth VSWR Noise Figure			1% for the group	
Control-Indicator (2A2) Coding Check	3.9	4.19.18	1%	
Control Monitor, Signal Level (2A4)	3.9	4.19.19		
VSWR Insertion Loss Isolation Measurement			1% for the group	
Transmitter, Radio (5TR1) Panel, Monitor (5TR1A1) Continuity Check	3.9	4.19.20	1%	
Synthesizer (5TR1A2)	3.9	4.19.5		
Frequency Check Synch. Alarm Deviation Power Output			1% for the group	
Control, Alarm (5TR1A3)	3.9	4.19.21	1%	
Low Power Alarm Backlash Adjustment				
Amplifier-Frequency Multiplier (5TR1A4)	3.9	4.19.22		

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Frequency Response Output Level Power Output (with ALC) Harmonic Suppression Input VSWR			1% for the group	
Amplifier-Monitor (5TR1A5)	3.9	4.19.23		
Operational Check Metering Check Frequency Response			1% for the group	
Power Supply (5TR1PS1)	3.9	4.19.24		
Insulation and Resistance Preliminary Check Meter Reading Input Rectification High Tension Rise Time and Frequency Measurement at 5 KHz Converter Rise Time and Frequency Measurement at 400 Hz Converter 26V dc Unreg. Supply Filament Supply 28V dc/12V dc Reg. Supply High Frequency Instability Constant Current High Tension Metering Circuit Calibration 5KHz Transistor Saturation Blower Voltage and Alarm Supply Hum Measurement (120 Hz) Spike Measurement (5KHz) Input AC Current Short Circuit Protection Check Starting Circuit Check			1% for the group	
Case, Transmitter (5A2) CY-4637( )/GRC-103(V)				

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Distribution Box (5A2A1) Continuity Check	3.9	4.19.25	1%	
Amplifier-Frequency Multiplier (6) AM-4320( )/GRC-103(V)				
Amplifier, Radio Frequency (6AR1)	3.9	4.19.26		
Drive Power Calibration				
Output Amplifier-Power Output			1% for the group	
Output Amplifier-Input Matching				
Driver Amplifier-Power Output				
Driver Amplifier-Input Matching				
Complete Amplifier-Tuning and Power Output				
Electronic Switch (6A1)	3.9	4.19.33		
Operational Check				
VSWR			1% for the group	
Insertion Loss				
Rejection Ratio				
Power Monitoring				
Frequency Multiplier Assembly (6A2)	3.9	4.19.27		
Tuning Procedure Band 1B			1%	
Tuning Procedure Band 1A				
Control-Indicator (6A3)	3.9	4.19.28		
Coding Check			1%	
Receiver-Transmitter Order Wire (9) RT-773( )/GRC-103(V)				
Panel, Distribution (9A1)	3.9	4.19.29		
Point-to-point Resistance			1%	

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Panel, Monitor (9A2)	3.9	4.19.30		
Point-to-point Resistance			1%	
Converter, Telephone Signal (9A3)	3.9	4.19.31		
Oscillator Frequency Signal Receiver Selectivity Signal Receiver Sensitivity			1% for the group	
Amplifier Assembly (9A4)	3.9	4.19.32		
Thru Paths Power Gain Frequency Response Harmonic Distortion Crosstalk			1% for the group	
Insert Paths Power Gain Frequency Response Harmonic Distortion Signal Compression			1% for the group	
Drop Paths Power Gain Frequency Response Harmonic Distortion Sidetone Level			1% for the group	
(c) Electrical (Main Assemblies) Receiver, Radio R-1329( )(P)/GRC-103(V) with Amplifier-Converter AM-4316( )/ GRC-103(V)1				
Demodulation	3.4.2.3, 3.4.3.8, 3.4.4.2	4.15	1% for the group	
Noise Figure	3.4.2.2	4.15		

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4.5.1 TABLE VIII (Cont'd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
IF Bandwidth	3.4.2.5	4.19.1		
Received Signal Monitoring	3.4.2.4	4.15	1%	
Receiver-Protection	3.4.2.6	4.19.2	for the	
Frequency Response	3.4.2.1	4.15	group	
VSWR (Duplexers)	3.4.3.11	4.19.15		
Transmitter, Radio T-983( )(P)/GRC-103(V) with Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V)				
Frequency Response	3.4.1.5	4.16		
Power Output	3.4.1.1	4.16	1%	
Power Dissipation	3.4.1.3	4.16	for the	
Setability	3.4.1.4	4.16	group	
Low Power Alarm	3.4.1.2	4.16		
Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V)				
Tone Generator	3.4.4.1.1	4.17	1%	
Deviation	3.4.4.3	4.19.3	for	
Sidetone	3.4.4.11	4.19.32	the	
Peak Limiting	3.4.4.5	4.19.32	group	
Antenna AS-1852( )/GRC-103(V)				
Symmetry	3.4.5.1	4.13		
Squint	3.4.5.2	4.13	1%	
Minor Lobes	3.4.5.3	4.13	for	
Gain	3.4.5.4	4.13	the	
VSWR	3.4.5.5	4.13	group	
Impedance	3.4.5.6	4.13		

4.5.1.1 Order of Inspection within Group A - Group A inspection shall be performed in an order satisfactory to the Government.

4.5.2 Group B Inspection - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table IX and to the special inspection levels of Table 1 of MIL-STD-105.

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TABLE IX

Group B Inspection

Inspection	Req Para	Insp Para	AQL (Major)
(a) Electrical (Main Assemblies) Receiver, Radio R-1329( )(P)/GRC-103(V) with Amplifier-Converter AM-4316( )/ GRC-103(V)			
Output Impedance	3.4.3.2	4.19.1	6.5% for
Squelch and Alarm	3.4.3.7.3	4.19.1	the group
Transmitter, Radio T-983( )(P)/GRC-103(V) with Amplifier-Frequency Multiplier AM-4320( )/GRC-103(V)			
Input Impedance	3.4.3.1	4.19.3	6.5% for
PCM Modulation	3.4.3.4	4.19.3	the group
(b) Electrical System Radio Set AN/GRC-103( )(P)/(V)1			
Frequency Response	3.4.3.3	4.18	
PCM Signal-to-Noise	3.4.3.5	4.18	
PCM Distortion	3.4.3.6	4.18	
Pulse Regeneration	3.4.3.7.1	4.18	
Timing Signal	3.4.3.7.2	4.18	6.5% for
Order Wire Distortion	3.4.4.8	4.18	the
Order Wire Signal-to-Noise	3.4.4.4	4.18	group
Power Requirement	3.8	4.18, 4.11	
Order Wire Frequency Response	3.4.4.6	4.22	
Order Wire Net Loss	3.4.4.10	4.22	
Duplexer	3.4.3.9	4.19.15	
Duplexer Discrimination	3.4.3.10	4.19.15	
(c) Mechanical Requirements			
Mechanical Interchangeability	3.10	4.21	6.5%

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4.5.2.1 Group B Sampling Plans - The Group B AQL shall be as shown in Table IX and the inspection level shall be S-3.

4.5.2.2 Order of Inspection within Group B - Group B inspection shall be performed in the order listed in Table IX.

4.5.3 Group C Inspection - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table X. Samples shall be selected in accordance with 4.5.3.1.

TABLE X

Group C Inspection

Inspection	Req Para	Insp Para
Sub Group I		
Mast Erection	3.5.1	4.14
Mast Operation	3.5.2	4.14
Transmitter Modulation	3.4.1.7	4.19.3
Duplexer Insertion Loss	3.4.3.9	4.19.15
Systems Level Stability	3.4.3.12	4.9.1
Shock, Bench Handling	3.11.4	4.7.3
Shock Test; Ballistic	3.11.4	4.7.2
Bounce Test	3.11.4	4.7.1
Sub Group II		
Low Temperature	3.11.1.1	4.7.4.1
High Temperature	3.11.1.2	4.7.4.2
Altitude	3.11.3	4.7.6
Humidity	3.11.2	4.7.5
Sub Group III		
Electromagnetic Interference Test	3.4.1.6, 3.4.2.7, 3.12	4.23

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4.5.3.1 Sampling for Group C Inspection -

4.5.3.1.1 Sub Group I - For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 100 units. For subsequent Group C inspection, one unit shall be selected at random from each 100 units or fraction thereof. This unit shall be subjected to all tests under Sub-Group I in the order specified.

4.5.3.1.2 Sub Group II - For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 100 units. For subsequent Group C inspection, one unit shall be selected at random for each 100 units or fraction thereof. This unit shall be subjected to all tests under Sub-Group II in the order specified.

4.5.3.1.3 Sub Group III - For this sub-group, one unit shall be selected from the first production lot. This shall cover Group C requirements for the first 500 units. For subsequent Group C inspection, one unit shall be selected at random for each 500 units or fraction thereof. This unit shall be subjected to the Electromagnetic interference test.

4.5.3.2 Order of Inspection within Group C - Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.3.3 Non Compliance - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of the failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative (QAR). After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective.

4.5.4 Reinspection of Conforming Group C Sample Units - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection only.

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4.5.5 Group D Inspection - This inspection shall consist of the test specified in Table XI, and shall be performed on units from lots which have been subjected to and met Group A and B inspection.

TABLE XI

Group D Inspection

Inspection	Req Para	Insp Para
Reliability	3.15	4.12

4.5.5.1 Sampling for Group D Inspection - Samples shall be selected at random in accordance with paragraph 4.5.5.

4.5.5.2 Group D Failures - Actions required relative to Group D failures shall be in accordance with procedures specified in MIL-STD-781.

4.5.5.3 Reinspection of Conforming Group D Sample Units - Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and sample units are subjected to and pass Group A inspection.

4.6 Preconditioning -

4.6.1 Bounce preconditioning - The unit, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection. All equipment delivered on the contract or order shall be subjected to the requirements of this paragraph. If an equipment fails any test because of the bounce preconditioning, the defect applies to that test.

4.6.2 Electrical preconditioning (Burn-in Debugging) -

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4.6.2.1 Sequence of Testing - Before subjecting equipments to this procedure, the contractor must have available objective evidence of compliance to Group A requirements.

4.6.2.2 Test Conditions - Every equipment shall be subjected to a 48 hour burn-in period. The last 24 hours of burn-in shall be failure free. The burn-in shall be conducted as if it were the beginning of a Production Reliability Test of Appendix I to this specification. The failures shall be recorded and analyzed, but shall not be considered relevant for computation of equipment MTBF.

4.6.2.3 Compliance - Units passing burn-in debugging shall be submitted to Group A, B, and C inspection in that order without benefit of repair, cleaning, realignment, or other changes, unless otherwise specified herein. If after successful completion of burn-in debugging, the equipment fails either the Group A or B tests, the equipment will be reworked and resubmitted to burn-in debugging.

4.7 Service Conditions - (Radio only: For antenna test procedures, see SM-A-763665 and for Mast see SM-A-696070.)

4.7.1 Bounce Test; class L units - The equipment shall be tested in accordance with Method 514, Procedure IX, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in Table VII. During Part 1, all connections (such as cable assemblies) shall be in place. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439. The adapter plate shall be free of cracks and broken welds, and all unused holes, except the 2-inch breather holes, shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate. Additional barriers used to safeguard personnel shall not restrain the fore-and-aft motion of the adapter plate against the backstop.

4.7.2 Shock Test; ballistic - The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The equipment, including shock mounts (if any), shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. Prior to testing, the equipment shall meet full specification performance for the measurements specified in Table VII.

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4.7.3 Shock Test; bench-handling - The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to and after testing, the equipment shall meet full specification performance for the measurements specified in Table VII.

4.7.4 Temperature Test -

4.7.4.1 Low Temperature - The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -65°F and the operating temperature shall be -35°F. Prior to, during, and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1.

4.7.4.2 High Temperature - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 125°F. Prior to, during, and after testing, the equipment shall meet full specification performance for the measurements specified in 4.9.1.

4.7.5 Humidity (moisture-resistance) test - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 4.9.1. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.6 Altitude (elevation) test - The equipment shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810, except the chamber internal pressure in step 2 shall be 282.0 mm of Hg (11.1 in. of Hg or 25,000 ft. above sea level). During and after testing the equipment shall meet full specification performance for the measurements specified in 4.9.2. The chamber may be returned to ambient atmospheric pressure for frequency changes.

4.8 Standard inspection conditions - Unless otherwise specified, all tests shall be performed at 115 volts ± 5 percent from a single-phase source operating between 47.5 and 63.0 Hz. Tests shall be conducted at room ambient temperature, pressure and humidity unless otherwise specified.

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4.9 Test specifications (service conditions) - The equipment shall be tested in accordance with the following test specifications and meet the requirements of this specification. The tests shall be performed during periods as specified in MIL-STD-810.

4.9.1 Temperature and humidity - 4.2 and 4.5 of SM-A-763739 except the test be performed at channels 40, 100, 160, 220, 310 and 400 instead of those specified. Tests at line frequencies of 47.5 Hz and 420 Hz and high and low line voltage shall be performed at any one channel. 4.1 of SM-A-763740 with the exception that tests be performed at channels 40, 100, 160, 220, 300 and 400 instead of those specified. 4.6 of SM-A-763739 with the exception that it be performed at channels 40, 160, 280 and 400 instead of those specified. 4.5 of SM-A-763738 except tests be performed at channels 50, 180 and 400 instead of those specified. 4.8 and 4.9 of SM-A-763738, except test be performed at channel 240 instead of those specified. 4.2.7 of SM-A-763738, 4.2 of SM-A-763740, except frequencies below 10 kHz be deleted from frequency response measurement. 4.5 of SM-A-763740, 4.6.8 of SM-A-763740.

4.9.2 Altitude - 4.2 of SM-A-763739, except that the test be performed at channels 40 and 340 instead of those specified. 4.5 of SM-A-763739, except the test be performed at channel 340 instead of the channel specified. 4.6 of SM-A-763739, except that the test be performed at channel 340 instead of the channel specified. 4.5 of SM-A-763738, except that the test be performed at channel 240 instead of the channel specified, 4.5 of SM-A-763740.

4.10 Operational test service conditions - With the equipment set up in a system, an operational test shall be performed. The test shall consist of voice communications, order wire ringing through the systems plus a visual check of the metering facilities for readings specified in 4.1f and 4.2.1a(f) of SM-A-763740.

4.11 A.C. Power line, Voltage and Frequency Tests - The following tests shall be performed at room ambient and at channels 40, 180 and 400. The tests shall be performed at 105 volts 47 Hz, 115 volts 47 Hz, 125 volts 47 Hz, 105 volts 420 Hz, 115 volts 420 Hz and 125 volts 420 Hz.

INSPECTION	REQUIREMENT PARA
4.2 and 4.5 of SM-A-763739	3.4.1
4.5 of SM-A-763738	3.4.2.2
4.2.5 of SM-A-698826	3.4.3.7
4.2.2 of SM-A-763740	3.4.3.4
4.10 of this specification	3.4.4.1

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4.12 Reliability Testing -

4.12.1 Reliability Qualification Test - The contractor shall perform a Reliability Qualification Test in accordance with Test Plan IV of MIL-STD-781 and Appendix I to this specification. The minimum test time per equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781.

4.12.2 Production Reliability Sampling - After successfully passing the Reliability Qualification Test, production reliability sampling shall begin. The Production Sampling Tests shall commence with the first lot available after the models used for first article testing. The samples shall be tested in accordance with Test Plan IVA of MIL-STD-781 and Appendix I to this specification. The specified (θ) MTBF shall be 1200 hours. The minimum test time per equipment shall be in accordance with paragraph 4.2.2.4 of MIL-STD-781.

4.12.3 Testing - All testing shall be conducted under the environment of Test Level B of MIL-STD-781. Equipment on-off cycling shall be on for 144 hours, off 8 hours. Vibration shall be for two minutes of every operating hour during one eight-hour period each week.

4.12.4 Failure Definition - A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification or the contractors approved test procedures. Paragraph 5.5 through 5.10 inclusive of MIL-STD-781 shall apply. The types of failures and required actions shall be in accordance with paragraph 5.5 of MIL-STD-781. Determination of compliance shall be in accordance with paragraph 5.4.8 of MIL-STD-781.

4.12.5 Rework and Retest Provisions - Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the lot unless positive action consisting of design change, modification, or rework actually relating to observed failures has been applied to the lot. Any subsequent rerun must start at zero hours operating time.

4.13 Antenna Test Procedure - The antenna shall be tested in accordance with SM-A-763665 and meet the requirements of 3.4.5.

4.14 Mast Test Procedures - The mast shall be tested in accordance with SM-A-696070 and meet the reqirements of 3.5.

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4.15 Radio Set AN/GRC-103( )(V)1 Receiver Section Test Procedure - The receiver section shall be tested in accordance with paragraphs 4.3, 4.5, 4.2 and 4.10 of SM-A-763738 and meet the requirements of 3.4.2.1; 3.4.2.2; 3.4.2.3, 3.4.3.8, and 3.4.4.2; and 3.4.2.4, respectively.

4.16 Radio Set AN/GRC-103( )(V)1 Transmitter Section Test Procedure - The transmitter section shall be tested in accordance with paragraphs 4.5, 4.2, 4.3, 4.4 and 4.7 of SM-A-763739 and meet the requirements of 3.4.1.4, 3.4.1.1, 3.4.1.2, 3.4.1.3 and 3.4.1.5, respectively.

4.17 Receiver-Transmitter, Order Wire RT-773( )/GRC-103(V) Test - The receiver-transmitter, order wire shall be tested in accordance with 4.2.5 of SM-A-698807 and meet the requirement of 3.4.4.1.1.

4.18 Radio Set AN/GRC-103( )(V)1 Test Procedure - The radio set shall be tested in accordance with paragraphs 4.2.3, 4.3, 4.4, 4.5, 4.6.7, 4.6.8, 4.7 and 4.8 of SM-A-763740 and meet the requirements of 3.4.3.3; 3.4.3.5; 3.4.3.6; 3.4.3.7.1 and 3.4.3.7.2; 3.4.4.8; 3.4.4.4; 3.4.2.6;and 3.8, respectively.

4.19 Component Test Procedures - The assemblies, subassemblies and modules of Radio Set AN/GRC-103( )/(V) shall be tested in accordance with the following drawings in order to demonstrate compliance with 3.9 of this specification.

4.19.1 Receiver, Radio R-1329( )(P)/GRC-103(V) - The Receiver, Radio R-1329( )(P)/GRC-103(V) shall be tested in accordance with SM-A-698826 and meet the requirements thereof. Meeting paragraph 4.2.15 of this drawing demonstrates compliance with 3.4.2.5. Tests for 3.4.3.2 and 3.4.3.7.3 are found in paragraphs 4.2.12 and 4.2.8 of SM-A-698826, respectively.

4.19.2 Amplifier-Converter, AM-4316/GRC-103(V) - The Amplifier-Converter AM-4316( )/GRC-103(V) shall be tested in accordance with SM-A-763700 and meet the requirements thereof.

4.19.3 Transmitter, Radio T-983( )(P)/GRC-103(V) - The Transmitter, Radio T-983( )(P)/GRC-103(V) shall be tested in accordance with SM-A-698831 and meet the requirements thereof. Tests for the requirements of paragraphs 3.4.4.3, 3.4.3.1, 3.4.3.4 and 3.4.1.7 are found in paragraphs 4.3.5, 4.9, 4.3.1 through 4.3.3 and 4.3.4 of SM-A-698831.

4.19.4 Amplifier Frequency Multiplier AM-4320/GRC-103(V) - The Amplifier Frequency Multiplier AM-4320( )/GRC-103(V) shall be tested in accordance with SM-A-763733 and meet the requirements thereof.

4.19.5 Synthesizer (IRE1A2/5TR1A2) - The synthesizer shall be tested in accordance with SM-A-698628 and meet the requirements thereof.

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4.19.6 Restorer, Pulse Form (IRE1A3) - The restorer, pulse form shall be tested in accordance with SM-A-698241 and meet the requirements thereof.

4.19.7 Limiter-Discriminator (IRE1A4) - The limiter-discriminator shall be tested in accordance with SM-A-698031 and meet the requirements thereof.

4.19.8 Panel Monitor (IRE1A1) - The panel, monitor shall be tested for continuity in accordance with SM-A-698317 and meet the requirements thereof.

4.19.9 Power Supply (IRE1PS1) - The power supply shall be tested in accordance with SM-A-698372 and meet the requirements thereof.

4.19.10 Amplifier-Frequency Multiplier (IRE1A5) - The amplifier-frequency multiplier shall be tested in accordance with SM-A-698610 and meet the requirements thereof.

4.19.11 Amplifier, Video (IRE1AR1) - The amplifier, video shall be tested in accordance with SM-A-698035 and meet the requirements thereof.

4.19.12 Amplifier, Intermediate (1RE1AR2) - The amplifier, intermediate shall be tested in accordance with SM-A-698002 and meet the requirements thereof.

4.19.13 Distribution Box (1A2) - The distribution box shall be tested for continuity in accordance with SM-A-698685 and meet the requirements thereof.

4.19.14 Tuner, Radio Frequency (2A1) - The tuner shall be tested in accordance with SM-A-763700 and meet the requirements thereof.

4.19.15 Duplexer (2A1A1) - The Duplexer shall be tested in accordance with SM-A-764157 and SM-A-763698 and meet the requirements thereof. Tests for the requirements of paragraphs 3.4.3.11, 3.4.3.9 and 3.4.3.10 are found in paragraphs 4.4.2; 4.2.2 and 4.2.3; and 4.3 of SM-A-764157, respectively.

4.19.16 Converter, Frequency, Electronic (2A1A2) - The converter, frequency, electronic shall be tested in accordance with SM-A-763695 and meet the requirements thereof.

4.19.17 Amplifier, Radio, Frequency (2A1AR1) - The amplifer, radio frequency shall be tested in accordance with SM-A-763697 and meet the requirements thereof.

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4.19.18 Control Indicator Coding Check (2A2) - The control indicator shall be tested in accordance with SM-A-763699 and meet the requirements thereof.

4.19.19 Control Monitor, Signal Level (2A4) - The control monitor shall be tested in accordance with SM-A-763688 and meet the requirements thereof.

4.19.20 Panel, Monitor (5TR1A1) - The panel monitor shall be tested in accordance with SM-A-698790 and meet the requirements thereof.

4.19.21 Control, Alarm (5TR1A3) - The control, alarm shall be tested in accordance with SM-A-698776 and meet the requirements thereof.

4.19.22 Amplifier-Frequency Multiplier (5TR1A4) - The amplifier-frequency multiplier shall be tested in accordance with SM-A-696443 and meet the requirements thereof.

4.19.23 Amplifier-Monitor (5TR1A5) - The amplifier-monitor shall be tested in accordance with SM-A-698664 and meet the requirements thereof.

4.19.24 Power Supply (5TR1PS1) - The power supply shall be tested in accordance with SM-A-698363 and meet the requirements thereof.

4.19.25 Distribution Box (5A2A1) - The distribution box shall be tested in accordance with SM-A-698789 and meet the requirements thereof.

4.19.26 Amplifier, Radio (6AR1) - The amplifier, radio shall be tested in accordance with SM-A-764122 and meet the requirements thereof.

4.19.27 Frequency Multiplier Assembly (6A2) - The frequency multiplier assembly shall be tested in accordance with SM-A-763731 and meet the requirements thereof.

4.19.28 Control-Indicator (6A3) - The control-indicator shall be tested in accordance with SM-A-763699 and meet the requirements thereof.

4.19.29 Panel, Distribution (9A1) - The panel, distribution shall be tested in accordance with SM-A-698759 and meet the requirements thereof.

4.19.30 Panel, Monitor (9A2) - The panel, monitor shall be tested in accordance with SM-A-698695 and meet the requirements thereof.

4.19.31 Converter, Telephone Signal (9A3) - The converter, telephone signal shall be tested in accordance with SM-A-698805 and meet the requirements thereof.

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4.19.32 Amplifier Assembly (9A4) - The amplifier assembly shall be tested in accordance with SM-A-698806 and meet the requirements thereof. Tests for the requirements of 3.4.4.11 and 3.4.4.5 are found in paragraphs 4.1.2, 4.1.5, 4.2.2, 4.3.2 and 4.3.5; and 4.2.5 of SM-A-698806.

4.19.33 Electronic Switch (6A1) - The electronic switch shall be tested in accordance with SM-A-763730 and meet the requirements thereof.

4.20 Visual and Mechanical Inspection - Parts and equipments shall be examined for the defects listed in MIL-STD-252 and the safety hazards of 3.10.

4.21 Mechanical Interchangeability - Mechanical interchangeability shall be demonstrated by gaging the applicable parts, assemblies and units with the gages as specified in the bid request and contract.

4.22 Order Wire Operational Tests - Two terminals shall be set up to operate in a full duplex PCM system connected through sufficient attenuation. Frequency response and net loss measurements shall be made with the equipment meeting the requirements of 3.4.4.6 and 3.4.4.10.

4.23 Electromagnetic Interference Tests - The AN/GRC-103( )(V)1 shall be tested for compliance to the electromagnetic interference requirements of 3.4.1.6, 3.4.2.7 and 3.12. All test methods and setups shall be in accordance with the requirements of MIL-STD-462 and Appendix of MIL-STD-461. Broadband emission tests shall be performed with the AN/GRC-103( )(V)1 operating at 310 MHz. All tests except CE06 shall be performed with the equipment operating in a full duplex mode.

4.24 Quality Conformance Inspection of Preparation for Delivery - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 The preparation for delivery provisions shall be in accordance with MIL-R-45292.

6. NOTES

6.1 Intended Use - The equipment covered by this specification is a complete, compact, rugged, radio relay set which may be used either as a terminal or modulation frequency patch repeater for intermediate or forward area systems. The radio set is used to provide simultaneous two-way (duplex) transmission. The system provides radio-frequency (rf) channel assignments from 220 MHz to 404.5 MHz. When used in combination, either as a repeater or as a terminal, it is possible to tune and operate the receiver and transmitter on independent frequencies over the frequency range.

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6.2 Ordering Data - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Levels of preservation, packaging and packing (see Section 5).

c. When rough handling is required.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature, tools and running spare parts.

g. First Article inspection (see 3.1).

h. Requirements for gages (see 4.21).

6.3 Group C Inspection - Approval to ship may be withheld at the discretion of the Government pending the decision from the Contracting Officer on the adequacy of corrective action (see 4.5.3.3).

6.3.1 Initial Shipments - Initial shipment of end items will be contingent only upon successful completion of all Group C tests (see 4.5.3).

6.4 Nomenclature - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GRC-103W(V)1. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Color - The color chip furnished by the procuring agency will be color chip No. X24087 and may be obtained upon request to Commander, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

Custodian:

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Preparing Activity

Army - (EL)

Project No. 5820-A618

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APPENDIX I

RELIABILITY TEST PROGRAM FOR RADIO SET AN/GRC-103 (V)1

10. SCOPE

10.1 The Reliability Test Program for the Radio Set AN/GRC-103(V)1 shall be completed as per the following test plans:

a. Reliability Qualification Test, in accordance with MIL-STD-781, Test Plan IV, on four (4) systems to be selected from the First Article equipments.

b. Production Reliability Sampling Test, in accordance with MIL-STD-781, Test Plan IV A, on four (4) production systems selected from each production lot. A lot shall be defined for reliability test purposes as twenty-five (25) units.

10.2 The Reliability Demonstration shall be completed after reaching the acceptance criteria of plan IV or IVA (as applicable) for MIL-STD-781.

The MTBF (Θo) for the above plans shall be 1200 hours for the system.

20. TEST CONDITIONS

20.1 The four equipments shall be set up in a two-way, two-hop system as shown in Figure 20.1 using suitable attenuators to absorb the output power from each transmitter and to provide an appropriate level of signal to each following receiver. The four equipments shall carry a P.C.M. signal and any deterioration in performance of any piece of equipment in the system shall produce error signals at the receiving end. The faulty equipment will then be located by checking individual meter readings.

20.2 The test shall be conducted in a suitable test chamber maintained at 40° ± 5°C. (95°F to 113°F).

20.3 The equipment shall be accessible to an operator for switching and retuning as required, and for interchanging of the R. F. Heads as called for in the test procedure.

20.4 The radio equipment shall operate continuously for periods of 144 hours and shall be switched off for periods of eight hours. During these "off periods" the R.F. Heads shall be interchanged as detailed in sub-section 30. A time recording meter shall be used to show total running time for the equipment.

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20.5 For two minutes of every operating hour during one eight-hour period each week, each equipment shall be vibrated at 2.2g ± 10% peak acceleration value at any non-resonant frequency between 20Hz and 60Hz, measured at the mounting points of the equipment.

20.6 The equipment undergoing reliability testing shall be subjected to a burn-in period of at least 48 hours prior to the start of reliability testing. This period of equipment operation shall be performed in accordance with applicable portions of the equipment specification.

30. TEST PROCEDURE

30.1 With the four radio sets connected as shown in Figure 20.1, set them to the first group of channels shown in Table 30.1.

30.2 With the P.C.M. test set on, adjust the transmitter No. A input control to provide ± 180 KHz peak deviation as indicated by the panel meter reading (in the green band when switched to 12 CH. P.C.M.).

30.3 Check the meter reading of the following receiver No. B when switched to 12 CH. P.C.M. It should show a reading in the green band.

30.4 Adjust the input control of transmitter No. C for ± 180KHz peak deviation as indicated on the panel meter and check the following receiver No. D.

30.5 Repeat for each equipment group in the "go" path and then back on the "return" path. Under normal conditions there shall be no appreciable error signal output from the P.C.M. test set and the electronic counter shall only give random indications. In the event of a fault developing anywhere in the system, error signals shall appear at the output and shall give visual indications on the counter. In order to provide continuous unattended monitoring, it is desirable that a digital recorder be connected to the output of the counter.

30.6 Every four hours throughout the test period, the operating channels shall be changed in accordance with the sequence in Table 30.1. Channel changes shall continue until an Accept/Reject decision is reached (see MIL-STD 781B, Tables IV and IVA).

30.7 Every six days the R.F. Tuning Heads shall be interchanged in accordance with the schedule in Table 30.2. This will be carried out during the off periods defined in sub-section 20. Tuning Heads shall be changed in the following manner:

a. The fixed heads shall be designated alphabetically and shall remain in fixed positions with the case. The R.F. heads shall be designated numerically and shall be moved as required throughout the demonstration.

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b. The identifying numbers are prefixed by the letters T or R to indicate whether they are transmitting or receiving heads.

c. If a fourth (4) change is required, Table 30.2 will revert back to the start position and the sequence shall begin again; this shall continue as long as changes are required.

30.8 The meter readings of all equipments shall be logged together with information on R.F. heads and channels used, after each frequency change. This log shall be in the form shown in Figure 30.1. Order wire signalling shall be checked daily between all equipments on test.

40. FAILURE ACTION

40.1. Only failures occurring within the equipment under test directly as a result of normal operation shall be considered as relevant. Failures in the following categories are classed as non-relevant:

a. Failures in any of the test equipment used.

b. Failures due to accidents or improper test procedures.

c. Failures occurring during the burn-in period.

d. Failures which can be eliminated by mutually agreed corrections.

e. Failures caused by other failures.

40.2 When a suspected relevant failure occurs in any equipment, the test shall be stopped and the failed equipment removed, repaired, and reinstalled at which time testing may continue.

40.3 The failed equipment shall be appropriately tagged to show the nature of the failure and the accumulated hours of operation. A tag shall be retained on the equipment and a notation of time of return to test added. Full details of the failure shall be recorded as detailed in sub-section 50 and the failure shall be recorded on the form shown in Figure 30.2.

50. TEST RECORDS

50.1 A complete operating log shall be kept during the test to record all pertinent information such as operating hours, off periods, vibration periods, channel changes, R. F. Head changes, meter readings, and times at which failures occur or when units are restored to test.

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50.2 In addition to this operating log, an equipment failure record shall be kept for every equipment under test in the format given in specification MIL-STD-781.

50.3 To assist in the demonstration of the M.T.B.F. for the equipment, a failure summary record shall be compiled as shown in MIL-STD-781 and shall include all failures classed as relevant on all equipments under test.

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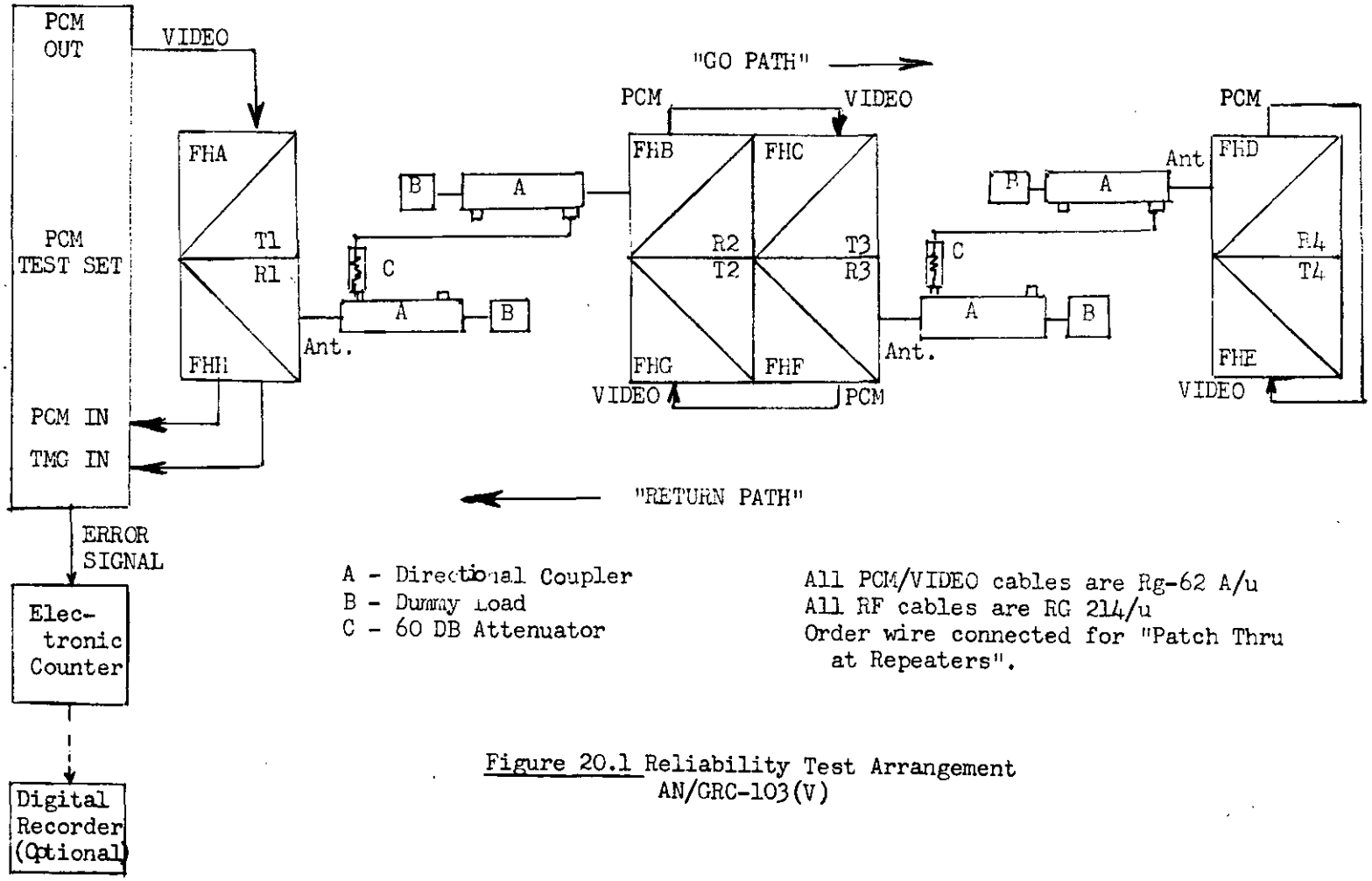
TABLE 20.1

LIST OF TEST EQUIPMENT

DESCRIPTION	MANUFACTURER	TYPE
Vibration Machine 0-60Hz	Lab Corporation	Model 30-300 or equivalent
Accelerometer	Columbia Labs	Model 1700 or equivalent
Random PCM Generator		USAECOM Dwg SM-E-698042 or equivalent
Electronic Counter	Hewlett-Packard	5245L or equivalent
Frequency Converter	Hewlett-Packard	5254C or equivalent
Directional Coupler	Hewlett-Packard	Model 778D or equivalent
Attenuator 60DB, 2W	Hewlett-Packard	8491A-660 or equivalent

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Normal Supply to All Equipments 115Vac 60Hz
Temperature Maintained at 40 ± 5°C (95 to 113°F)



A - Directional Coupler
B - Dummy Load
C - 60 DB Attenuator

All PCM/VIDEO cables are Rg-62 A/u
All RF cables are RG 214/u
Order wire connected for "Patch Thru at Repeaters".

Figure 20.1 Reliability Test Arrangement AN/GRC-103(V)

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TABLE 30.1

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
0	40	217	66	243
4	234	57	260	83
8	74	100	100	277
12	268	91	294	117
16	108	285	134	311
20	302	125	328	151
24	142	319	168	345
28	336	159	362	185
32	176	353	202	379
36	370	193	396	219
40	210	387	43	220
44	404	227	237	60
48	51	228	77	254
52	245	68	271	94
56	85	262	111	288
60	279	102	305	128
64	119	296	145	322
68	313	130	339	162
72	153	330	179	356
76	347	170	373	196
80	187	364	213	390
84	381	204	407	230
88	221	398	54	231
92	222	45	248	71
96	62	239	88	265
100	256	79	282	105
104	96	273	122	299
108	290	113	316	139
112	130	307	156	333
116	324	147	350	173

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TABLE 30.1(CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
120	164	341	190	367
124	358	181	384	207
128	198	375	224	401
132	392	215	225	48
136	232	409	65	242
140	233	56	259	82
144	73	250	99	276
148	267	90	293	116
152	107	284	133	310
156	301	124	327	150
160	141	318	167	344
164	335	158	361	184
168	175	352	201	378
172	369	192	395	218
176	209	386	42	219
180	403	226	236	59
184	50	227	76	253
188	244	67	270	93
192	84	261	110	287
196	278	101	304	127
200	118	295	144	321
204	312	135	338	161
208	152	329	178	355
212	346	169	372	195
216	186	363	212	389
220	380	203	406	229
224	220	397	53	230
228	221	44	247	70
232	61	238	87	264
236	255	78	281	104

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
240	95	272	121	298
244	289	112	315	138
248	129	306	155	332
252	323	146	349	172
256	163	340	189	366
260	357	180	383	206
264	197	374	223	400
268	391	214	224	47
272	231	408	64	241
276	232	55	258	81
280	72	249	98	275
284	266	89	292	115
288	106	283	132	309
292	300	123	326	149
296	140	317	166	343
300	334	157	360	183
304	174	351	200	377
308	386	191	394	217
312	208	385	41	218
316	402	225	235	58
320	49	226	75	252
324	243	66	269	92
328	83	260	109	286
332	277	100	303	126
336	117	294	143	320
340	311	134	337	160
344	151	328	177	354
348	345	168	371	194
352	185	362	211	388
356	379	202	405	228

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS			
	T1	T2	T3	T4
	R2	R1	R4	R3
360	219	396	52	229
364	220	43	246	69
368	60	237	86	263
372	254	77	280	103
376	94	271	120	297
380	288	111	314	137
384	128	305	154	331
388	322	145	348	171
392	162	339	188	365
396	356	179	382	205
400	196	373	222	399
404	390	213	223	46
408	230	407	63	240
412	231	54	257	80
416	71	248	97	274
420	265	88	291	114
424	105	282	131	308
428	299	122	325	148
432	139	316	165	342
436	333	156	359	182
440	173	350	199	376
444	367	190	393	216
448	207	384	40	217
452	401	224	234	57
456	48	225	74	251
460	242	65	263	91
464	82	259	108	285
468	276	99	302	125
472	116	293	142	319
476	310	133	336	159

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
480	150	327	176	353
484	344	167	370	193
488	184	361	210	387
492	378	201	404	227
496	218	395	51	228
500	219	42	245	68
504	59	236	85	262
508	253	76	279	102
512	93	270	119	296
516	287	110	313	136
520	127	304	153	330
524	321	144	347	170
528	161	338	187	364
532	355	178	381	204
536	195	372	221	398
540	389	212	222	45
544	229	406	62	239
548	230	53	256	79
552	70	247	96	273
556	264	87	290	113
560	104	281	130	307
564	298	121	324	147
568	138	315	164	341
572	352	155	358	181
576	172	349	198	375
580	366	189	392	215
584	206	383	232	409
588	400	223	233	56
592	47	224	73	250
596	241	64	267	90

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
600	81	258	107	284
604	275	98	301	124
608	115	292	141	318
612	309	132	335	158
616	149	326	175	352
620	343	166	369	192
624	183	360	209	386
628	377	200	403	226
632	217	394	50	227
636	218	41	244	67
640	58	235	84	261
644	252	75	278	101
648	92	269	118	295
652	286	109	312	135
656	126	303	152	329
600	320	143	346	169
664	160	337	186	363
668	354	177	380	203
672	194	371	220	397
676	388	211	221	44
680	228	405	61	238
684	229	52	255	78
688	69	246	95	272
692	263	86	289	112
696	103	280	129	306
700	297	120	323	146
704	137	314	163	340
708	331	154	357	180
712	171	348	197	374
716	365	188	391	214

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
720	205	382	231	408
724	399	222	232	55
728	46	223	72	249
732	240	63	266	89
736	80	257	106	283
740	274	97	300	123
744	114	291	140	317
748	308	131	334	157
752	148	325	174	351
756	342	165	368	191
760	182	359	208	385
764	376	199	402	225
768	216	393	49	226
772	217	40	243	66
776	57	234	83	260
780	251	74	277	100
784	91	268	117	294
788	285	108	311	134
792	125	302	151	328
796	319	142	345	168
800	159	336	185	362
804	353	176	379	202
808	193	370	219	396
812	387	210	220	43
816	227	404	60	237
820	228	51	254	77
824	68	245	94	271
828	262	85	288	111
832	102	279	128	305
836	296	119	322	145

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS			
	T1	T2	T3	T4
	R2	R1	R4	R3
840	136	313	162	339
844	330	153	356	179
848	170	347	196	373
852	364	187	390	213
856	204	381	230	407
860	398	221	231	54
864	45	222	71	248
868	239	62	265	88
872	79	256	105	282
876	273	96	299	122
880	113	290	139	316
884	307	130	333	156
888	147	324	173	350
892	341	164	367	190
896	181	358	207	384
900	375	198	401	224
904	215	392	48	225
908	409	232	242	65
912	56	233	82	259
916	250	73	276	99
920	90	267	116	293
924	284	107	310	133
928	124	301	150	327
932	318	141	344	167
936	158	335	184	361
940	352	175	378	201
944	192	369	218	395
948	386	209	219	42
952	226	403	59	236
956	227	50	253	76

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 R2	T2 R1	T3 R4	T4 R3
960	67	244	93	270
964	261	84	287	110
968	101	278	127	304
972	295	118	321	144
976	135	312	161	338
980	329	152	355	178
984	169	346	195	372
988	363	186	389	212
992	203	380	229	406
996	397	220	230	53
1000	44	221	70	247
1004	238	61	264	87
1008	78	255	104	281
1012	272	95	298	121
1016	112	289	138	315
1020	306	129	332	155
1024	146	323	172	349
1028	340	163	366	189
1032	180	357	206	383
1036	374	197	400	223
1040	214	391	47	224
1044	408	231	241	64
1048	55	232	81	258
1052	249	72	275	98
1056	89	266	115	292
1060	283	106	309	132
1064	123	300	149	326
1068	317	140	343	166
1072	157	334	183	360
1076	351	174	377	200

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS T1 T2 T3 T4 R2 R1 R4 R3			
1080	191	368	217	394
1084	385	208	218	41
1088	225	402	58	235
1092	226	49	252	75
1096	66	243	92	269
1100	260	83	286	109
1104	100	277	126	303
1108	294	117	320	143
1112	134	311	160	337
1116	328	151	354	177
1120	168	345	194	371
1124	362	185	388	211
1128	202	379	228	405
1132	396	219	229	52
1136	43	220	69	246
1140	237	60	263	86
1144	77	254	103	280
1148	271	94	297	120
1152	111	288	137	314
1156	305	128	331	154
1160	145	322	171	348
1164	339	162	365	188
1168	179	356	205	382
1172	373	196	399	222
1176	213	390	46	223
1180	407	230	240	63
1184	54	231	80	257
1188	248	71	274	97
1192	88	265	114	291
1196	282	105	308	131

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TABLE 30.1 (CONT'D)

ELAPSED HOURS OF OPERATING TIME	RF TUNING HEAD CHANGE PAIRS			
	T1	T2	T3	T4
	R2	R1	R4	R3
1200	122	299	148	325
1204	316	139	342	165
1208	156	333	182	359
1212	350	173	376	199
1216	190	367	216	393
1220	384	207	217	40
1224	224	401	57	234

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RX FIXED HEAD

DATE	TIME	RF HEAD NO	CH NO	RECEIVER METER READINGS (F.S.D.)											O.W. CHECK
				180 KHz 12CH	300 KHz 24CH	OW	+12V DC	-12V DC	REFL PWR	XMTR DUPL	OSC	DOUB	MULT	RCVR SIG	
LIMITS				40-60%	40-60%	40-60%	40-60%	40-60%	less than 15%	above 40%	20-90%	20-90%	above 25%	30-90%	Light on Buzzer on (1600 Hz)

TX FIXED HEAD

DATE	TIME	RF HEAD NO	CH NO	TRANSMITTER METER READINGS (F.S.D.)										O.W. CHECK
				180 KHz 12CH	300 KHz 24CH	12V DC	28V DC	500V DC	OSC	DOUB	MULT	DRIV	PWR OUT	REFL PWR	
LIMITS				40-60%	40-60%	40-60%	40-60%	40-60%	20-90%	20-90%	20-90%	20-90%	above 40%	less than 15%	Light on Buzzer on (1600 Hz)

FIGURE 30.1

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TABLE 30.2

SCHEDULE FOR INTERCHANGING OF R.F. TUNING HEADS

R.F. HEAD IDENT NBRS.	Fixed Head Identifying Letters				
	START	FIRST CHANGE	SECOND CHANGE	THIRD CHANGE	FOURTH CHANGE
T1	A	G	C	E	A
R2	B	F	D	H	B
T3	C	E	A	G	C
R4	D	H	B	F	D
T4	E	A	G	C	E
R3	F	D	H	B	F
T2	G	C	E	A	G
R1	H	B	F	D	H

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FIGURE 30.2 FAILURE DATA SHEET TECHNICIAN _____

PART 1 - TO BE COMPLETED AT TIME OF FAILURE

1. F.H./R.F. HEAD COMBINATION _____ /_____
2. DAYS/HOURS FROM START _____ /_____
3. CHANNEL AT TIME OF FAILURE CH. _____
4. SYMPTOMS OF FAILURE: _____

_____

5. UNIT REMOVED FROM TEST (F.H. OR R.F. HEAD) _____
6. FAILURE REPORT FORM (F.R.F.) NO. _____
7. TOTAL DOWN-TIME OF FAILED UNIT _____

PART 2 - TO BE COMPLETED WHEN F.R.F. IS CLOSED

1. SUB-UNIT FAILED _____
2. DWG. NO. _____
3. SERIAL NO. _____
4. COMPONENT REPLACED _____
5. COMMENTS: _____

_____

Signature _____

Date _____

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Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-R-49023(EL) RADIO SET AN/GRC-103(V)1

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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